

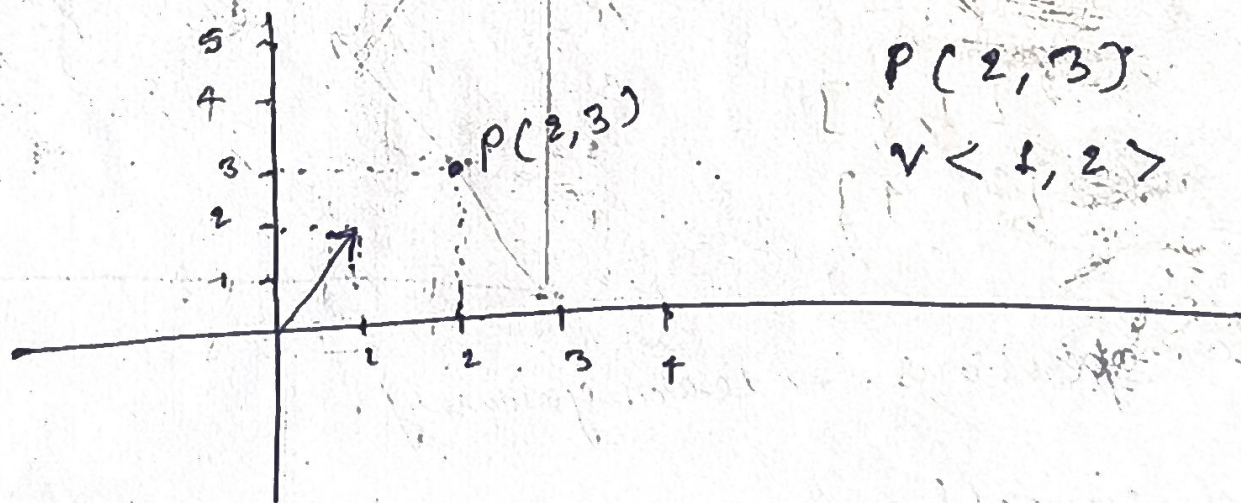
Algebra: It is one of the broad area of mathematics. In its most general form, algebra is the study of mathematical symbol and the rule for manipulating these symbol.

Linear Algebra: It is study of system of linear equation or linear

Scalar: A quantity that has magnitude but no particular direction is described by scalar.

Vector: A quantity that has magnitude and acts in a particular direction is described as vector.

Point: A point has position in space. The only characteristic that distinguishes one point from another is its position.



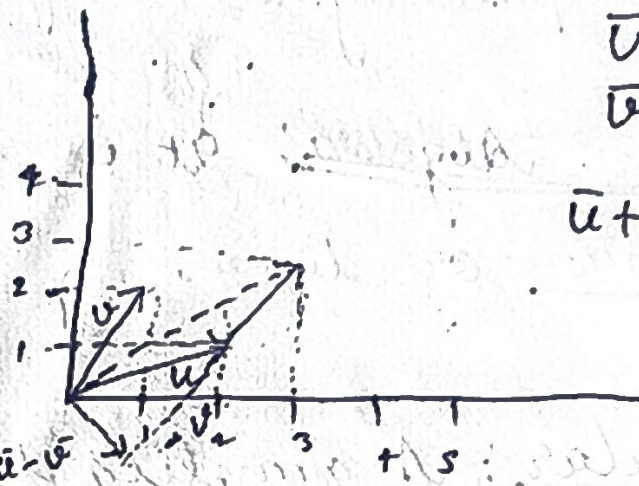
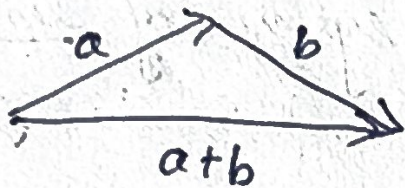
vector operation

↳ Addition

↳ Subtraction

∴ ↳ scalar multiplication

Addition:

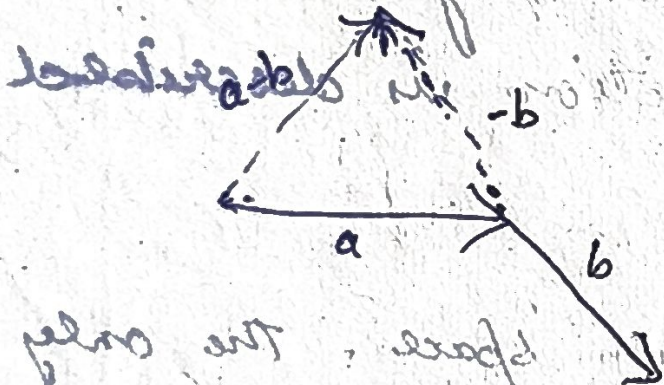


$$u = [2, 1]$$

$$v = [1, 2]$$

$$u+v = [3, 3]$$

Subtraction:

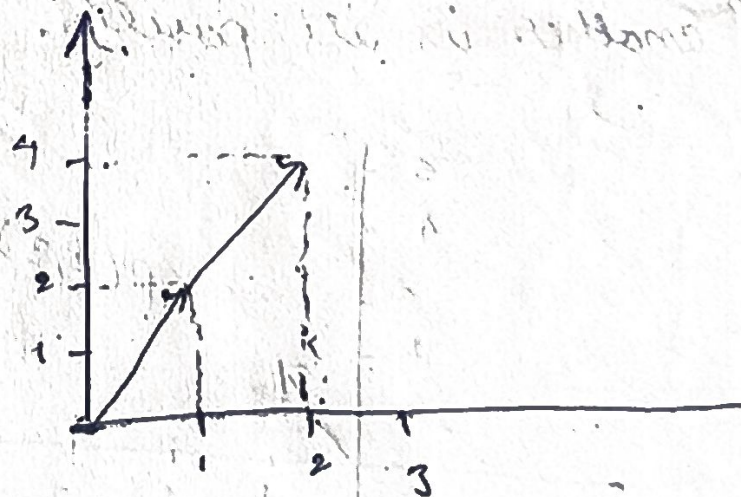


Multiplication

$$u = [1, 2]$$

$$= 2 [1, 2]$$

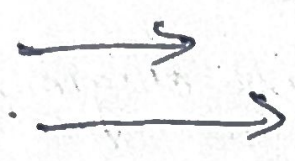
$$= [2, 4]$$



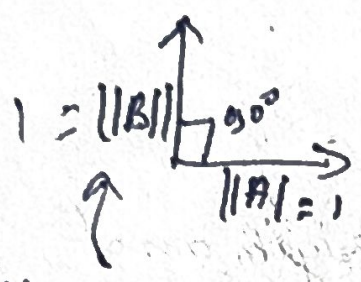
vector operations. → learn yourself.

Note: Read yourself vector norm, dot product, cross product.

Parallel vector

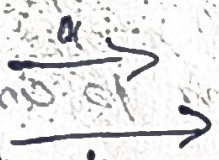


Perpendicular vector (orthogonal vector)

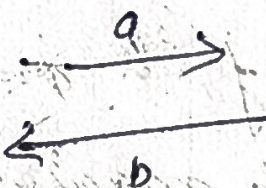


Orthogonal vector

Like and unlike vector

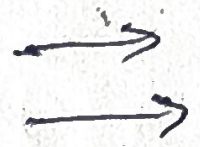


parallel with like vectors



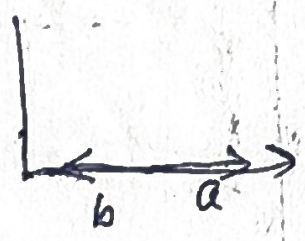
unlike vector

Equal vector



dirⁿ and magnitude are same.

Collinear vector



same plane with opposite dirⁿ.

Unit vector

$$|\hat{A}| = 1 \quad \hat{A}$$

Position vector



Start with origin.

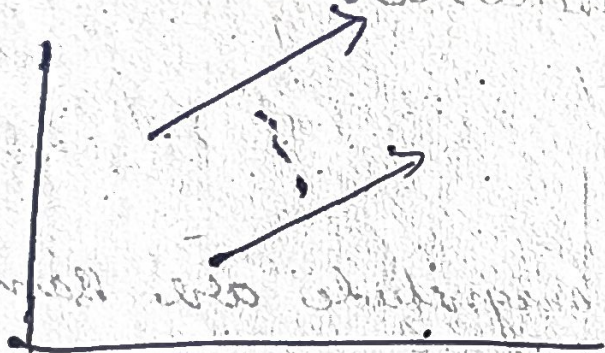
Zero vector



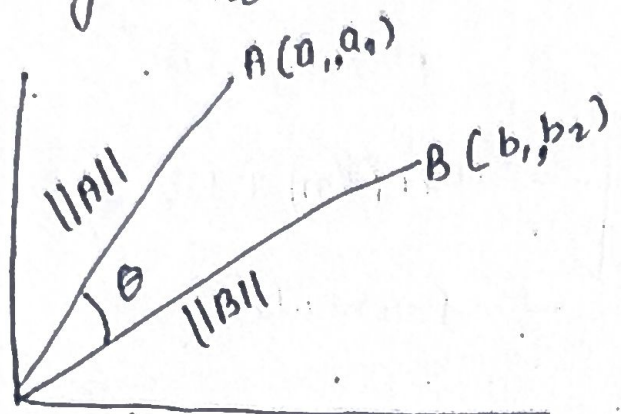
Start with origin and
end at the origin
K/a zero
vector

Free vector:

Place a vector from one position to another
but their dirⁿ & magnitude are same



Angle b/w 2 vectors:



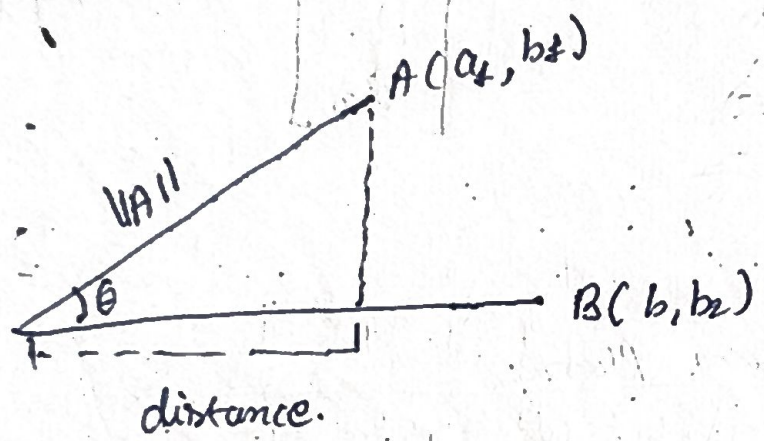
$$A \cdot B = ||A|| \times ||B|| \times \cos \theta$$

or

$$\theta = \cos^{-1} \left\{ \frac{a_1 \times b_1 + a_2 \times b_2}{||A|| ||B||} \right\}$$

$$\theta = \cos^{-1} \left\{ \frac{\sum_{i=1}^n a_i \times b_i}{||A|| ||B||} \right\}$$

Projection



$$\text{distance} = ||A|| \cos \theta$$

$$A \cdot B = ||A|| \times ||B|| \times \cos \theta$$

$$A \cdot B = ||B|| \times \text{distance}$$

$$\therefore \text{distance} = \frac{A \cdot B}{||B||}$$

Unit vector:

$$\hat{a} = \frac{a}{||a||}$$

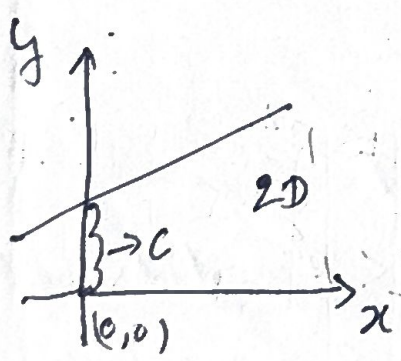
$$a = [3, 4]$$

$$||a|| = \sqrt{3^2 + 4^2} = 5$$

$$\hat{a} = \frac{a}{||a||} = \frac{3}{5}, \frac{4}{5}$$

$$\therefore ||\hat{a}|| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$$

Line vs Plane.

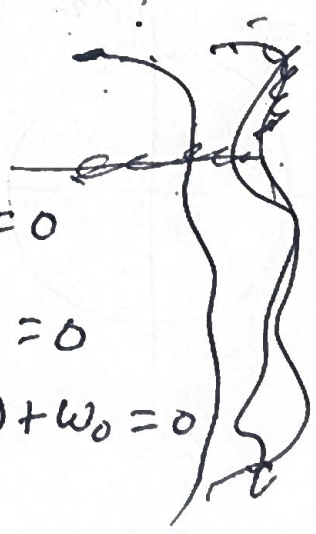


$$y = mx + c$$

$$A(x) + B(y) + C = 0$$

$$A(x_1) + B(x_2) + C = 0$$

$$w_1(x_1) + w_2(x_2) + w_0 = 0$$



2D line equation.

Plane

$$w_1(x_1) + w_2(x_2) + w_3(x_3) + w_0 = 0 \quad \text{Plane equation}$$

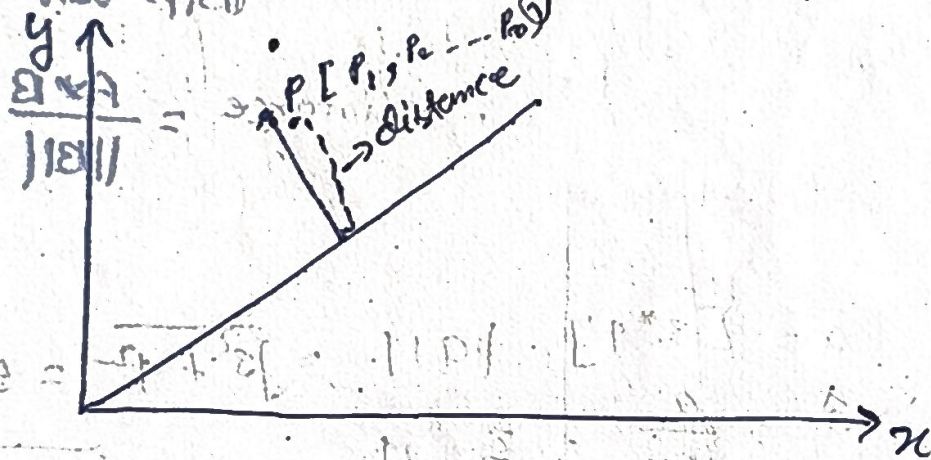
$$w_1(x_1) + w_2(x_2) + w_3(x_3) + \dots + w_n(x_n) + w_0 = 0$$

$$\therefore w_0 + \sum_{i=1}^n w_i(x_i) = 0 \quad \rightarrow \text{formula}$$

$$w_0 + [w_1, w_2, \dots, w_n] \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = 0 \quad \rightarrow \text{Vector Representation}$$

$$w_0 + w^T X = 0$$

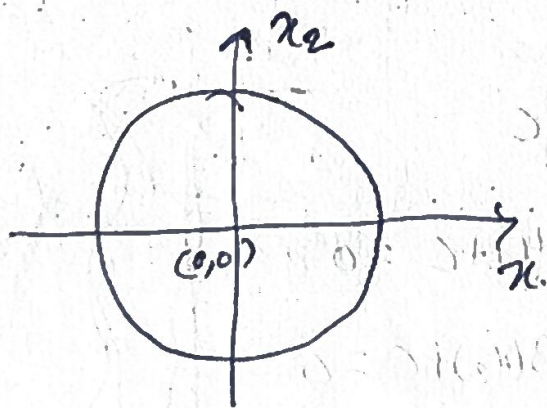
Distance of a Point from Plane:



$w^T X = 0$ eqnⁿ of line which pass through origin.

$$\text{distance} = \frac{w^T P}{\|w\|}$$

Circle



$$x^2 + y^2 = R^2$$

$$x^2 + y^2 < R^2 \rightarrow \text{inside the circle}$$

$$x^2 + y^2 > R^2 \rightarrow \text{outside " "}$$

$$x_1^2 + x_2^2 = R^2$$

linear algebra.

Distance formula:
$$\sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

3-D

$$= \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

Matrices (NumPy)

vector

Vector:

↳ Row vector: $A = [a_1, a_2, a_3, a_4, a_5]_{1,5}$ (list of elements)

↳ Column vector: $A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}_{5,1}$ (n, 1)

2D Matrix (list of list):

$$A = \begin{bmatrix} 1 & 5 & 9 \\ 2 & 6 & 10 \\ 3 & 7 & 11 \\ 4 & 8 & 12 \end{bmatrix} \quad \begin{matrix} \uparrow \\ c \end{matrix}$$

(Row x column)
4x3
(m x n)

$$A_{ij} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \\ A_{41} & A_{42} & A_{43} \end{bmatrix}$$

Matrix as location

Transpose of matrix:

$$A = [a_1, a_2, a_3, a_4, a_5] \quad A^T = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$$

Similarly:

$$A = \begin{bmatrix} 1 & 5 & 9 \\ 2 & 6 & 10 \\ 3 & 7 & 11 \\ 4 & 8 & 12 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

Matrix addition:

$$A = [a_1, a_2, a_3, a_4, a_5]$$

$$B = [b_1, b_2, b_3, b_4, b_5]$$

vector addition.

same dimension

$$A+B = [a_1+b_1, a_2+b_2, a_3+b_3, a_4+b_4, a_5+b_5]$$

$$\begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix}$$

same dimension we can add matrix otherwise not

dot product.

$$A = [a_1, a_2, a_3, a_4, a_5]_{1 \times 5} * B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}_{5 \times 1}$$

$$\Rightarrow c = [a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3 + a_4 \times b_4 + a_5 \times b_5]$$

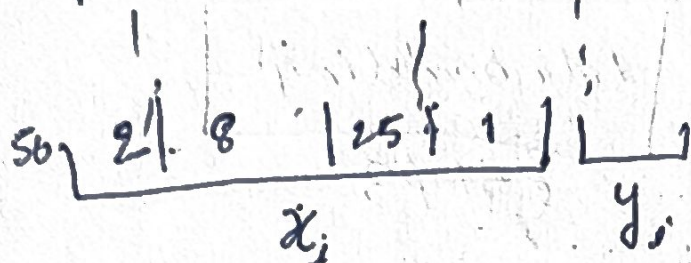
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Identity matrix

Diagonal matrix

Date Representation:

	f_1	f_2	f_3	f_4	q_p
1	2	5	10	3	1
2	5	10	30	2	0



$$\mathcal{D} = \{x_i, y_i\}_{i=1}^n$$

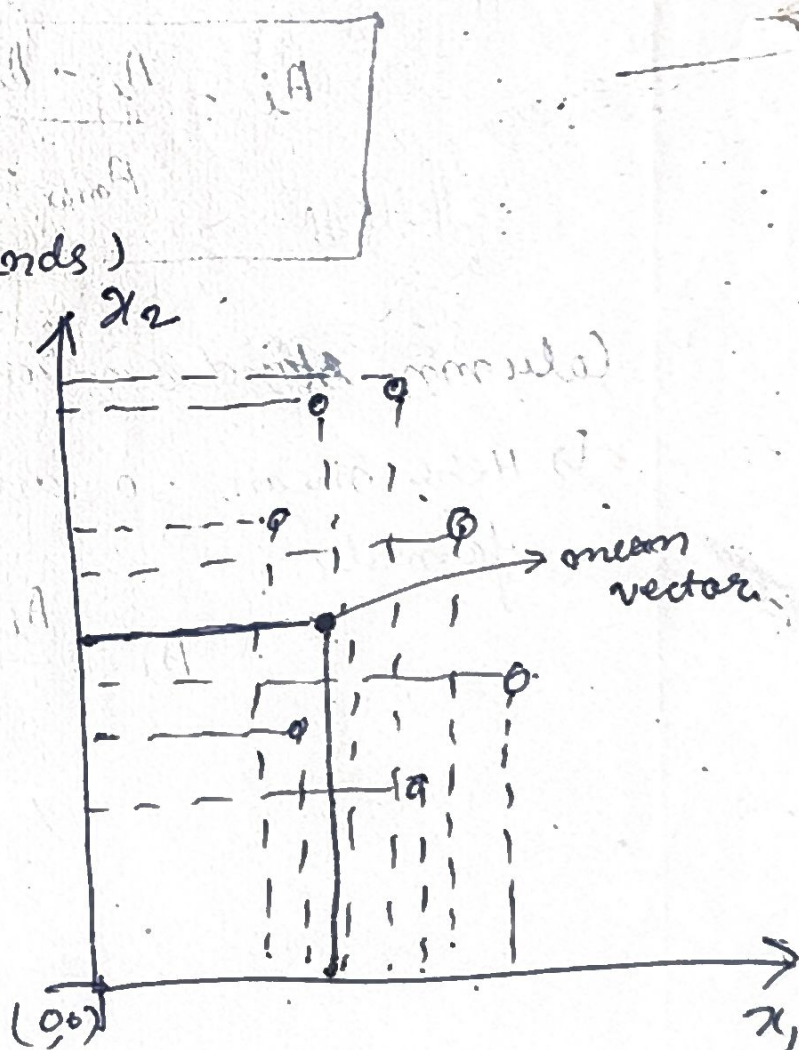
$$x_i \in \mathbb{R}^d, d=4$$

$$y_i \in \{1, 0\}$$

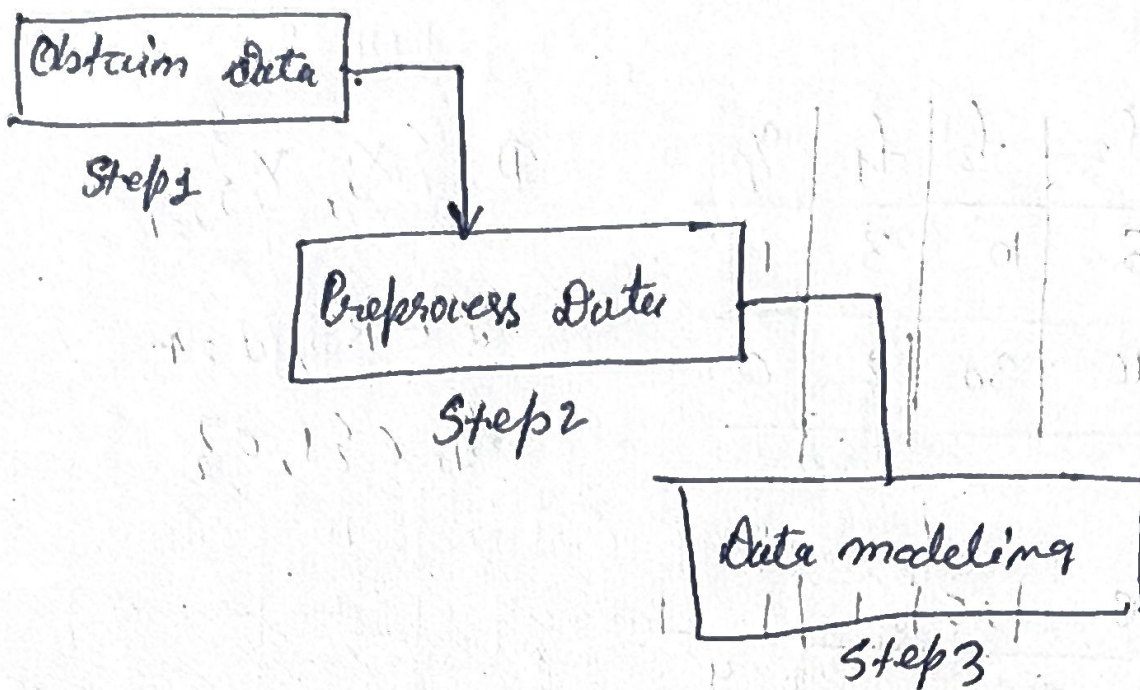
$$X = \begin{bmatrix} 2 & 5 & 10 & 3 \\ 5 & 10 & 30 & 2 \\ \vdots & \vdots & \vdots & \vdots \\ 2 & 8 & 25 & 1 \end{bmatrix}$$

Mean Vector

Player	x_1 Height (in)	x_2 Weight (pounds)
p_1	76	225
p_2	75	195
p_3	72	180
p_4	82	231
p_5	69	185
p_6	74	196
p_7	75	228
p_8	71	200
p_9	75	230



Data Preprocessing.



Step of Data preprocessing

Column normalization:

- ↳ Previous table are normalized in b/w 0 to 1
- ↳ Thus all column

$$A_i = \frac{A_i - A_{\min}}{A_{\max} - A_{\min}} \rightarrow \text{use this formula for normalization}$$

Column standardization

- ↳ Here mean = 0 and std-dev = 1 using following formula

$$A_i = \frac{A_i - \bar{A}}{\text{std.}}$$